For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T\to\infty}\text{var}(b_2|\mathbf{x}) = 0$.

a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \ldots, T$. Note that $\mathbf{x} = (1, 2, \ldots, T)$, $\sum_{t=1}^{T}(t-\bar{t})^2 = \sum_{t=1}^{T} t^2 - \left(\sum_{t=1}^{T} t\right)^2 / T$, $\sum_{t=1}^{T} t = T(T+1)/2$ and $\sum_{t=1}^{T} t^2 = T(T+1)(2T+1)/6$.

b. $y_t = \beta_1 + \beta_2(0.5)^t + e_t, t = 1, 2, \ldots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \ldots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \cdots + r^n = \frac{r(1-r^n)}{1-r}$$

c. Provide an intuitive explanation for these results.

a $\quad \text{Var}(b_2|X) = \dfrac{\sigma^2}{\Sigma(t-\bar{t})^2}$

$$\sum_{t=1}^{T}(t-\bar{t})^2 = \sum_{t=1}^{T} t^2 - \frac{(\sum_{t=1}^{T} t)^2}{T} = \frac{T(T+1)(2T+1)}{12} - \frac{(\frac{T(T+1)}{2})^2}{T} = \frac{2T(T+1)(2T+1)}{12} - \frac{3T(T+1)^2}{12}$$

$$= T(T+1)\frac{4T+2-3T-3}{12} = T(T+1)\frac{(T-1)}{12} = \frac{T(T^2-1)}{12}$$

$\text{Var}(b_2|X) = \dfrac{12\sigma^2}{T(T^2-1)} \Rightarrow \lim_{T\to\infty} \dfrac{12\sigma^2}{T(T^2-1)} = 0 \Rightarrow$ the least square estimator b_2 is consistent

b

$$\sum_{t=1}^{T}(x_t-\bar{x})^2 = \sum_{t=1}^{T} x_t^2 - \frac{(\Sigma x_t)^2}{T} = \frac{(1-(\frac{1}{4})^T)}{3} - \frac{(1-(\frac{1}{2})^T)^2}{T}$$

$$\begin{cases} \Sigma x_t = \sum_{t=1}^{T}(\frac{1}{2})^t = \frac{\frac{1}{2}(1-(\frac{1}{2})^T)}{1-\frac{1}{2}} = 1-(\frac{1}{2})^T \\ \Sigma x_t^2 = \sum_{t=1}^{T}(\frac{1}{4})^t = \frac{\frac{1}{4}(1-(\frac{1}{4})^T)}{1-\frac{1}{4}} = \frac{(1-(\frac{1}{4})^T)}{3} \end{cases}$$

$\text{var}(b_2|X) = \dfrac{\sigma^2}{\frac{(1-(\frac{1}{4})^T)}{3} - \frac{(1-(\frac{1}{2})^T)^2}{T}}$ $\quad \lim_{T\to\infty} \text{Var}(b_2|X) = 3\sigma^2 \neq 0 \Rightarrow$ least squares estimator b_2 is not consistent

c In (a) the x_t's become more dispersed as T increases, so OLS learns the slope precisely

In (b) the x_t's quickly bunch near 0, the extra data add little information, OLS can't estimate slope consistently

Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \qquad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

a. $\beta_2 = 0$
b. $\beta_1 + 2\beta_2 = 5$
c. $\beta_1 - \beta_2 + \beta_3 = 4$

a. $H_0: \beta_2 = 0$

$\text{Var}(b_2) = 4$

$b_2 = 3$

$t = \dfrac{3-0}{2} = 1.5$

$df = 63-3 = 60$

$t(0.975, 60) \approx 2$

$\because 1.5 < 2$

$\Rightarrow$ Do not reject H_0

b. $H_0: \beta_1 + 2\beta_2 = 5$

$\text{Var}(b_1+2b_2) = \text{Var}(b_1) + 4\text{Var}(b_2) + 4\text{cov}(b_1, b_2)$

$= 3 + 4\cdot 4 + 4\cdot(-2) = 11$

$se(b_1+2b_2) = \sqrt{11}$

$t = \dfrac{8-5}{\sqrt{11}} \approx 0.9$

$t(0.975, 60) \approx 2$

$\because 0.9 < 2$

$\Rightarrow$ Do not reject H_0

C. $\beta_1 - \beta_2 + \beta_3 = 4$

$\text{Var}(b_1 - b_2 + b_3) = \text{Var}(b_1) + \text{Var}(b_2) + \text{Var}(b_3) - 2\text{cov}(b_1, b_2) + 2\text{cov}(b_2, b_3) + 2\text{cov}(b_1, b_3)$

$= 3+4+3+4+1 = 15$

$se(b_1 - b_2 + b_3) = \sqrt{15}$

$t(0.975, 60) \approx 2$

$t = \dfrac{-2-4}{\sqrt{15}} \approx -1.55$

$\because 1.55 < 2$

$\Rightarrow$ Do not reject H_0

A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i$, $i = 1, 2, \ldots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x. Given that assumptions MR1–MR6 hold:

a. Show that $\hat{\beta}_{2,mean}$ is unbiased.

b. Derive an expression for $\text{var}\left(\hat{\beta}_{2,mean}|\mathbf{x}\right)$.

c. Write down an expression for $\text{var}\left(\hat{\beta}_{2,mean}\right)$.

d. Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2?

e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:

  i. $\text{var}(b_2|\mathbf{x})$ and $\text{var}\left(\hat{\beta}_{2,mean}|\mathbf{x}\right)$ where b_2 is the OLS estimator.

  ii. Estimates for $E\left[\left(s_x^2\right)^{-1}\right]$ and $E\left[4/(\bar{x}_2 - \bar{x}_1)^2\right]$ where s_x^2 is the sample standard deviation for x using N as a divisor.

f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?

g. Show that $E\left[\left(s_x^2\right)^{-1}\right] \xrightarrow{p} 0.12$ and $E\left[4/(\bar{x}_2 - \bar{x}_1)^2\right] \xrightarrow{p} 0.16$. [*Hint*: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]

h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

a. NTS: $\hat{\beta}_{2,mean}$ is unbiased

$\bar{y}_1 = \beta_1 + \beta_2 \bar{X}_1 + \bar{e}_1$ $\qquad$ $\bar{y}_2 - \bar{y}_1 = \beta_2(\bar{X}_2 - \bar{X}_1) + (\bar{e}_2 - \bar{e}_1)$

$\bar{y}_2 = \beta_1 + \beta_2 \bar{X}_2 + \bar{e}_2$ $\qquad$ $\hat{\beta}_{2\,mean} = \dfrac{\beta_2(\bar{X}_2 - \bar{X}_1) + (\bar{e}_2 - \bar{e}_1)}{\bar{X}_2 - \bar{X}_1} = \beta_2 + \dfrac{(\bar{e}_2 - \bar{e}_1)}{\bar{X}_2 - \bar{X}_1}$

By MR4, $E(e_i|X) = 0$, then $E(\bar{e}_2 - \bar{e}_1|X) = 0$

$\therefore E(\hat{\beta}_{2,mean}|X) = \beta_2$, which is unbiased

b. $Var(\hat{\beta}_{2,mean}|X) = Var\left(\dfrac{\bar{e}_2 - \bar{e}_1}{\bar{X}_2 - \bar{X}_1} \Big| X\right) = \dfrac{1}{(\bar{X}_2 - \bar{X}_1)^2} Var(\bar{e}_2 - \bar{e}_1|X) = \dfrac{1}{(\bar{X}_2 - \bar{X}_1)^2} \cdot \dfrac{4\sigma^2}{N} = \dfrac{4\sigma^2}{N(\bar{X}_2 - \bar{X}_1)^2}$

$$\left\{\begin{array}{l} \text{By MR5, } Var(e_i|X) = \sigma^2 \\ \text{Then } var(\bar{e}_i|X) = Var\left(\frac{2}{N}\Sigma e_i|X\right) = \left[\frac{2}{N}\right]^2\left(\frac{N}{2}\right)\sigma^2 = \frac{2\sigma^2}{N} \\ Var(\bar{e}_2 - \bar{e}_1|X) = var(\bar{e}_1|X) + Var(\bar{e}_2|X) = \frac{2\sigma^2}{N} + \frac{2\sigma^2}{N} = \frac{4\sigma^2}{N} \end{array}\right.$$

c.

$Var(\hat{\beta}_{2,mean}) = E[Var(\hat{\beta}_{2,mean}|X)] + Var[E(\hat{\beta}_{2,mean}|X)]$

$= E\left[\dfrac{4\sigma^2}{N(\bar{X}_2 - \bar{X}_1)^2}\right] + 0 = \dfrac{4\sigma^2}{N} E\left[\dfrac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right]$

d. NTS: $\hat{\beta}_{2\,mean} \xrightarrow{P} \beta_2$ when $N \to \infty$

if $Var(\hat{\beta}_{2\,mean}) \to 0$, then $\dfrac{1}{N} E\left[\dfrac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right] \to 0$ — ①

if $\bar{X}_2 - \bar{X}_1 \xrightarrow{P} C$ where $C \neq 0$ then $E\left[\dfrac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right] \to \dfrac{1}{C^2}$ — ②

$\because$ By ① & ②, under $\bar{X}_2 - \bar{X}_1 \xrightarrow{P} C$ where $C \neq 0$, $\dfrac{1}{N} E\left[\dfrac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right] \to 0$ as $N \to \infty$

$\therefore Var(\hat{\beta}_{2,mean}) \to 0$, then $\hat{\beta}_{2\,mean} \xrightarrow{P} \beta_2$

e.

N	$Var(b_2\|X)$	$var(\hat{\beta}_{2,mean}\|X)$	$E[(S_x^2)^{-1}]$	$E[4/(\bar{x}_2 - \bar{x}_1)^2]$
100	1.221	1.6493	0.1221	0.1647
500	0.241	0.3222	0.1205	0.1612
1000	0.1201	0.1604	0.1202	0.1604
5000	0.024	0.032	0.1200	0.1601

f. As N increase, $var(b_2|X)$, $var(\hat{\beta}_{2,mean}|X)$ decrease approximately $\frac{1}{N}$

$var(\hat{\beta}_{2,mean}|X)$ larger. $\because E[4/(\bar{x}_2-\bar{x}_1)^2] \to 0.16$, $E[(s_x^2)^{-1}] \to 0.12$

for larger N, $E[var(\hat{\beta}_{2,mean}|X)] \approx \frac{0.16\sigma^2}{N}$, $E[var(b_2, X)] \approx \frac{0.12\sigma^2}{N}$

OLS is more efficient but both $\to 0$ as $N \to \infty \Rightarrow \hat{\beta}_{2,mean}$ is consistant

g. Show that $E\left[(s_x^2)^{-1}\right] \xrightarrow{p} 0.12$ and $E\left[4/(\bar{x}_2-\bar{x}_1)^2\right] \xrightarrow{p} 0.16$. [*Hint*: The variance of a uniform random variable defined on the interval (a, b) is $(b-a)^2/12$.]

h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

g. NTS: $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$

$\because x \sim U(0,10)$

$\therefore var(x) = \frac{(b-a)^2}{12} = \frac{25}{3}$

$\because s_x^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})^2$

$\therefore s_x^2 \xrightarrow{p} var(x)$

$\therefore (s_x^2)^{-1} \xrightarrow{p} (\frac{25}{3})^{-1}$

$\therefore E[(s_x^2)^{-1}] \to \frac{12}{100} = 0.12$

NTS: $E[4/(\bar{x}_2-\bar{x}_1)^2] \xrightarrow{p} 0.16$

$\because \bar{x}_2 \xrightarrow{p} 7.5$, $\bar{x}_1 \xrightarrow{p} 2.5$

$\therefore \bar{x}_2-\bar{x}_1 \xrightarrow{p} 5$

$\therefore \frac{4}{(\bar{x}_2-\bar{x}_1)^2} \xrightarrow{p} \frac{4}{25}$

$\therefore E[4/(\bar{x}_2-\bar{x}_1)^2] \xrightarrow{p} 0.16$

h. No, not consistent

if $\bar{x}_2-\bar{x}_1 \to 0$, $\hat{\beta}_{2,mean}$ will be unstable, which means that $\hat{\beta}_{2,mean} \not\xrightarrow{p} \beta_2$